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# TELECOM CUSTOMER CHURN ANALYSIS (EDA)
# Author: Pratik
# Project Type: Exploratory Data Analysis (EDA)
# Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn
# Objective:
# Analyze customer churn behavior and identify key factors
# contributing to customer attrition.
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# =====
# 1. IMPORT LIBRARIES
# =====

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# 2. LOAD DATASET
# =====

df = pd.read_csv("Customer Churn.csv")

# Display first 10 records
df.head(10)

# Dataset dimensions (rows, columns)
df.shape

# Dataset information
df.info()

# =====
# 3. DATA CLEANING
# =====

# Replace blank values in TotalCharges with 0
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")

# Convert TotalCharges to float datatype
df["TotalCharges"] = df["TotalCharges"].astype(float)

# Check missing values
df.isnull().sum()

# Check duplicate rows

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df.duplicated().sum()

# Check duplicate Customer IDs
df["customerID"].duplicated().sum()

# =====
# 4. FEATURE TRANSFORMATION
# =====

# Convert SeniorCitizen column
# 1 = yes
# 0 = no

def conv(value):
    if value == 1:
        return "yes"
    else:
        return "no"

df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)

# Convert datatype to string
df["SeniorCitizen"] = df["SeniorCitizen"].astype(str)

df.head()

# =====
# 5. CHURN DISTRIBUTION ANALYSIS
# =====

# Count of customers who churned vs retained
a = sns.countplot(x="Churn", data=df)
a.bar_label(a.containers[0])
plt.title("Count of Customers by Churn")
plt.savefig(
    "Count of Customers by Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 6. CHURN PERCENTAGE ANALYSIS

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# =====
# Percentage share of churned and retained customers

gb = df.groupby("Churn").size()

plt.pie(
    gb,
    labels=gb.index,
    autopct="%1.2f%%"
)

plt.title("Percentage of Customers by Churn")

plt.savefig(
    "Percentage of Customers by Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 7. GENDER VS CHURN ANALYSIS
# =====

a = sns.countplot(
    x="gender",
    hue="Churn",
    data=df
)

a.bar_label(a.containers[0])
a.bar_label(a.containers[1])

plt.title("Count of Customers by Gender and Churn")

plt.savefig(
    "Count of Customers by Gender and Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 8. SENIOR CITIZEN VS CHURN ANALYSIS
# =====

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# Percentage churn comparison for senior and non-senior customers
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```
pct = pd.crosstab(  
    df["SeniorCitizen"],  
    df["Churn"],  
    normalize="index"  
) * 100
```

```
ax = pct.plot(  
    kind="bar",  
    stacked=True,  
    figsize=(8, 4)  
)
```

```
for container in ax.containers:  
    ax.bar_label(  
        container,  
        fmt="%1f%%",  
        label_type="center"  
    )
```

```
plt.title("Churn Percentage by Senior Citizen Status")  
plt.xlabel("Senior Citizen")  
plt.ylabel("Percentage")  
plt.legend(title="Churn")  
plt.xticks(rotation=0)
```

```
plt.savefig(  
    "Churn Percentage by Senior Citizen Status.png",  
    dpi=300,  
    bbox_inches="tight"  
)
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plt.show()
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# =====  
# 9. SENIOR CITIZEN DISTRIBUTION  
# =====
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```
gb = df.groupby("SeniorCitizen").size()
```

```
plt.pie(  
    gb,  
    labels=gb.index,  
    autopct="%1.2f%%"  
)
```

```
plt.title("Percentage of Customers by SeniorCitizen")
```

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plt.savefig(
    "Percentage of Customers by SeniorCitizen.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 10. TENURE VS CHURN ANALYSIS
# =====

plt.figure(figsize=(9, 4))

sns.histplot(
    x="tenure",
    hue="Churn",
    data=df,
    bins=72
)

plt.title("Count of Customers by Tenure and Churn")

plt.savefig(
    "Count of Customers by Tenure and Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 11. CONTRACT TYPE VS CHURN ANALYSIS
# =====

a = sns.countplot(
    x="Contract",
    data=df,
    hue="Churn"
)

a.bar_label(a.containers[0])

plt.title("Count of Customers by Contract")

plt.savefig(
    "Count of Customers by contract.png",
    dpi=300,
    bbox_inches="tight"
)

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)

plt.show()

# =====
# 12. SERVICE USAGE VS CHURN DASHBOARD
# =====

service_cols = [
    'PhoneService',
    'MultipleLines',
    'InternetService',
    'OnlineSecurity',
    'OnlineBackup',
    'DeviceProtection',
    'TechSupport',
    'StreamingTV',
    'StreamingMovies',
    'Contract',
    'PaperlessBilling',
    'PaymentMethod'
]

rows = 6
cols = 2

fig, axes = plt.subplots(
    rows,
    cols,
    figsize=(25, 30)
)

axes = axes.flatten()

for i, col in enumerate(service_cols):
    ax = sns.countplot(
        x=col,
        hue="Churn",
        data=df,
        ax=axes[i]
    )

    for container in ax.containers:
        ax.bar_label(container)

    ax.set_title(f"{col} vs Churn")
    ax.set_xlabel("")
    ax.set_ylabel("Count")

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    ax.tick_params(axis="x", rotation=0)
for j in range(len(service_cols), len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()

plt.savefig(
    "service_vs_churn_dashboard.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 13. MONTHLY CHARGES VS CHURN
# =====

plt.figure(figsize=(8, 5))

ax = sns.boxplot(
    x="Churn",
    y="MonthlyCharges",
    data=df
)

means = df.groupby("Churn")["MonthlyCharges"].mean()

for i, value in enumerate(means):
    plt.text(
        i,
        value,
        f"Mean={value:.2f}",
        ha="center",
        color="red"
    )

plt.title("Monthly Charges by Churn")

plt.savefig(
    "Monthly Charges by Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

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# 14. TOTAL CHARGES VS CHURN
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# =====
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plt.figure(figsize=(8, 5))

ax = sns.boxplot(
    x="Churn",
    y="TotalCharges",
    data=df
)

means = df.groupby("Churn")["TotalCharges"].mean()

for i, value in enumerate(means):
    plt.text(
        i,
        value,
        f"Mean={value:.2f}",
        ha="center",
        color="red",
        fontsize=12
    )

plt.title("Total Charges by Churn")

plt.savefig(
    "Total Charges by Churn.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
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# 15. CORRELATION ANALYSIS
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corr = df.select_dtypes(include=np.number).corr()

plt.figure(figsize=(8, 6))

sns.heatmap(
    corr,
    annot=True,
    cmap="rocket"
)

plt.title("Correlation Matrix")

plt.savefig(
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    "Correlation Matrix.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

# =====
# 16. DOWNLOAD GENERATED FILES
# =====

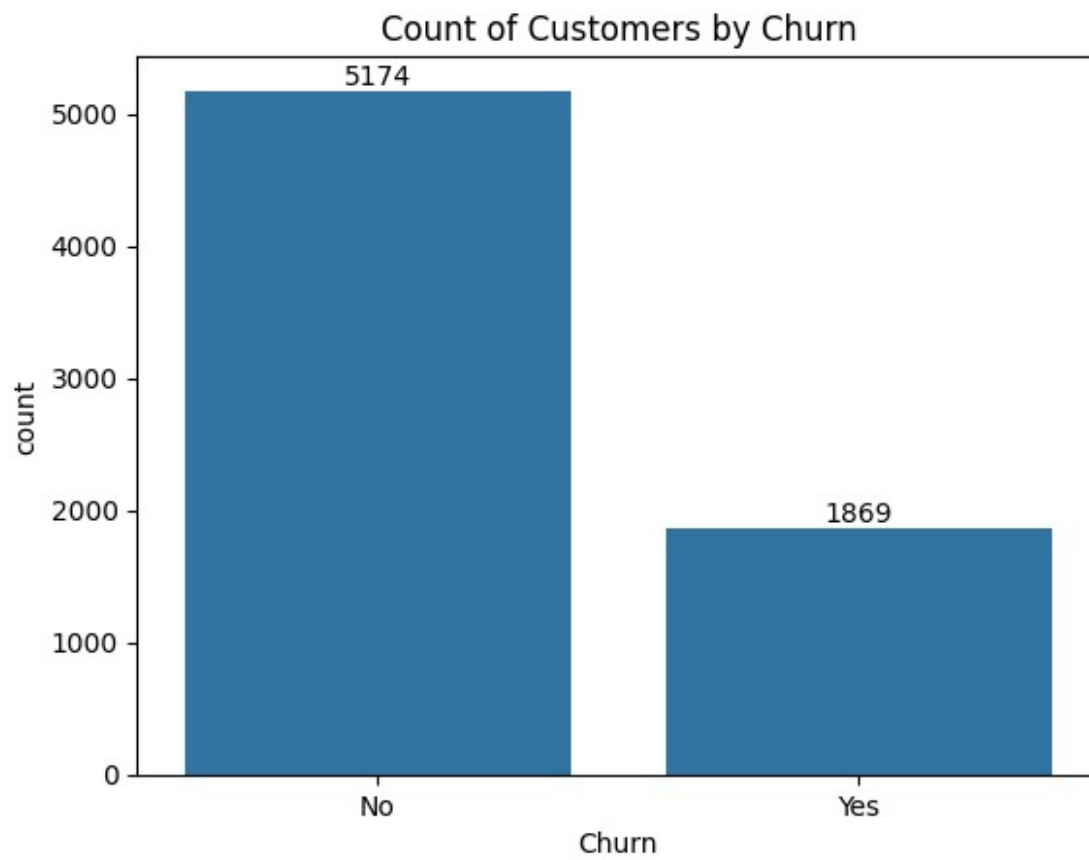
from google.colab import files

files.download("service_vs_churn_dashboard.png")
files.download("Count of Customers by contract.png")
files.download("Count of Customers by Tenure and Churn.png")
files.download("Percentage of Customers by SeniorCitizen.png")
files.download("Count of Customers by Gender and Churn.png")
files.download("Percentage of Customers by Churn.png")
files.download("Count of Customers by Churn.png")
files.download("Monthly Charges by Churn.png")
files.download("Total Charges by Churn.png")
files.download("Correlation Matrix.png")

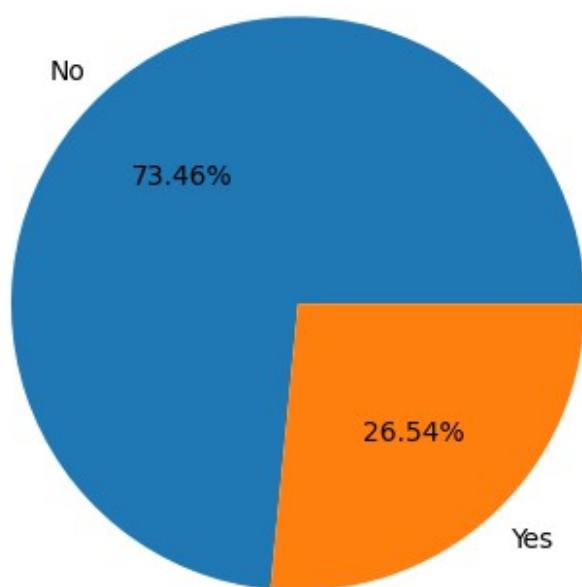
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null  object
1   gender                 7043 non-null  object
2   SeniorCitizen          7043 non-null  int64
3   Partner                7043 non-null  object
4   Dependents             7043 non-null  object
5   tenure                 7043 non-null  int64
6   PhoneService           7043 non-null  object
7   MultipleLines           7043 non-null  object
8   InternetService        7043 non-null  object
9   OnlineSecurity         7043 non-null  object
10  OnlineBackup            7043 non-null  object
11  DeviceProtection       7043 non-null  object
12  TechSupport            7043 non-null  object
13  StreamingTV            7043 non-null  object
14  StreamingMovies         7043 non-null  object
15  Contract               7043 non-null  object
16  PaperlessBilling        7043 non-null  object
17  PaymentMethod          7043 non-null  object
18  MonthlyCharges         7043 non-null  float64
19  TotalCharges           7043 non-null  object

```

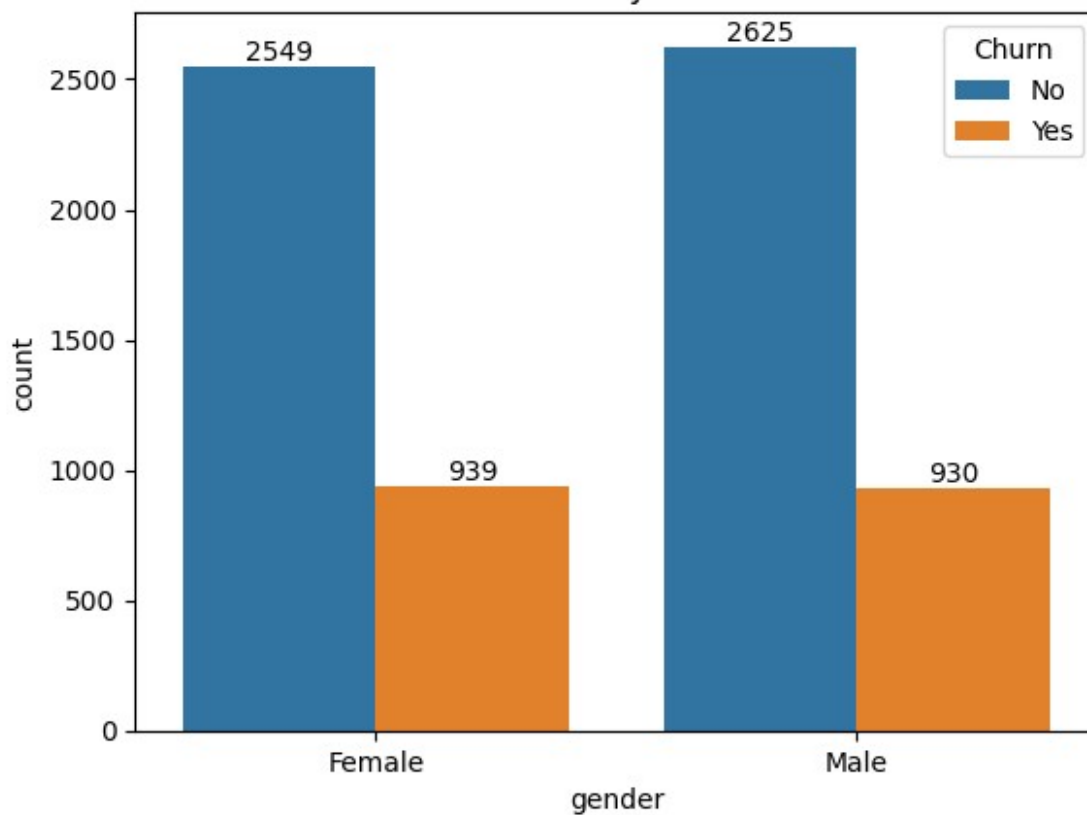
```
20  Churn          7043 non-null  object  
dtypes: float64(1), int64(2), object(18)  
memory usage: 1.1+ MB
```

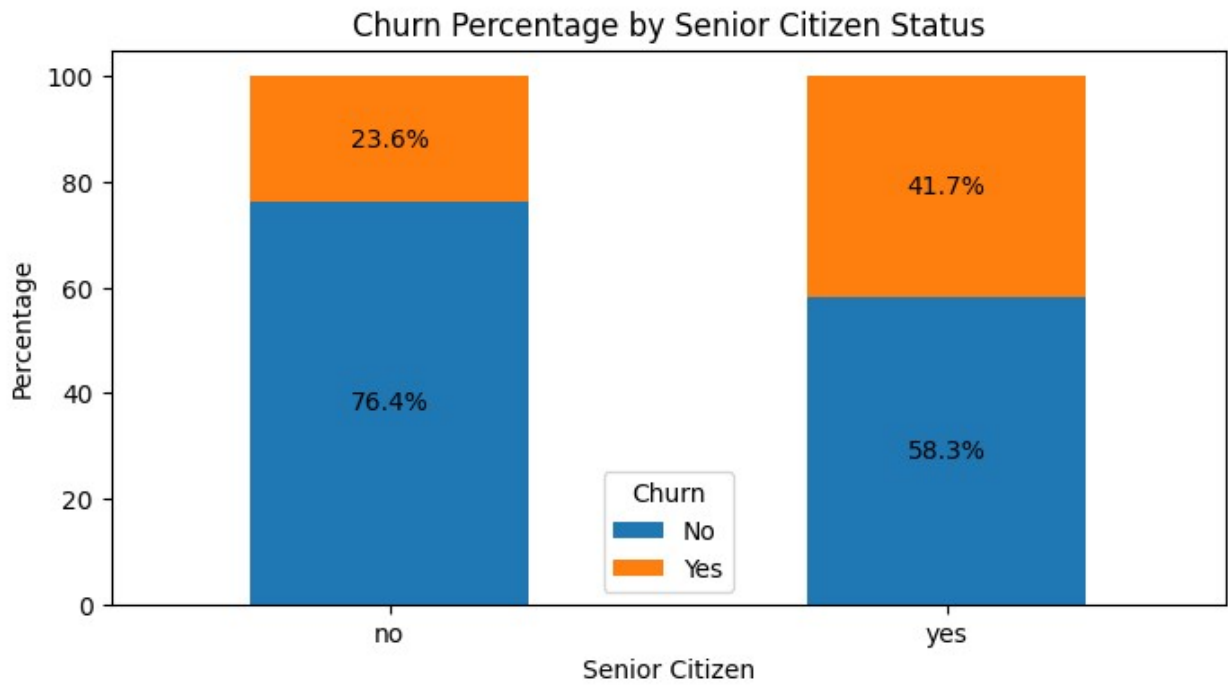


Percentage of Customers by Churn

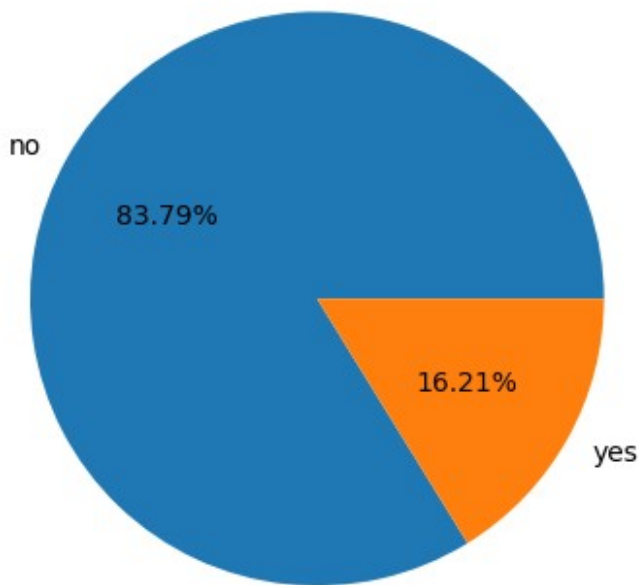


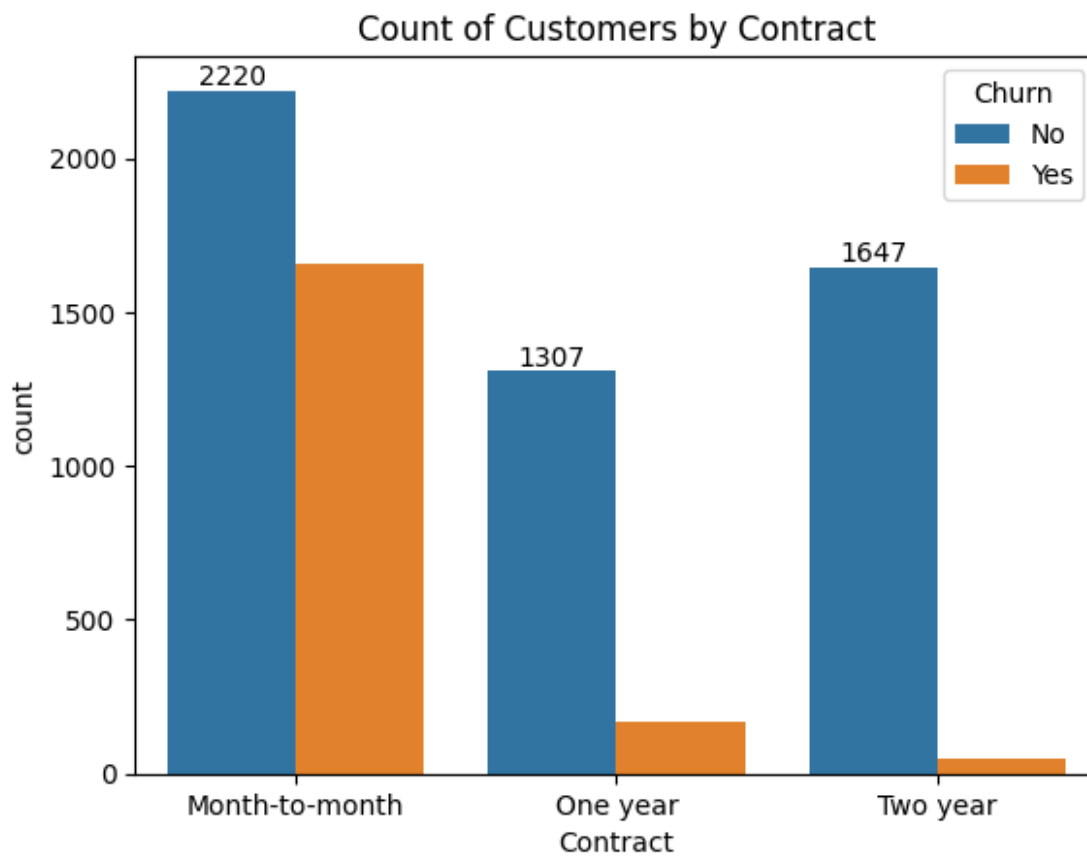
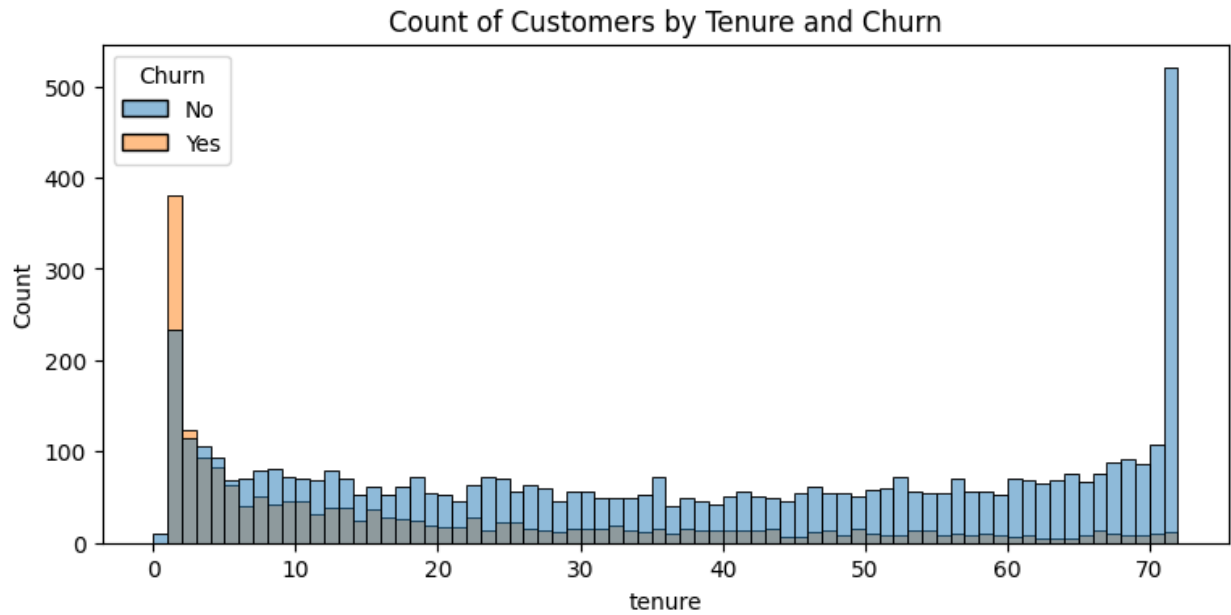
Count of Customers by Gender and Churn

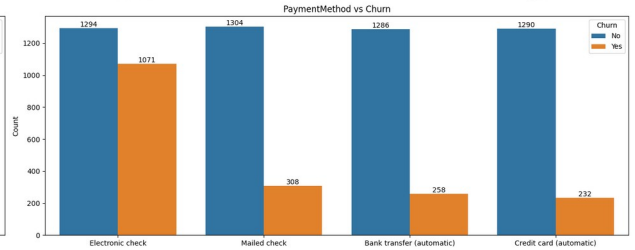
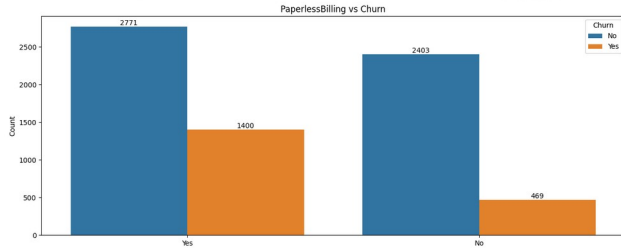
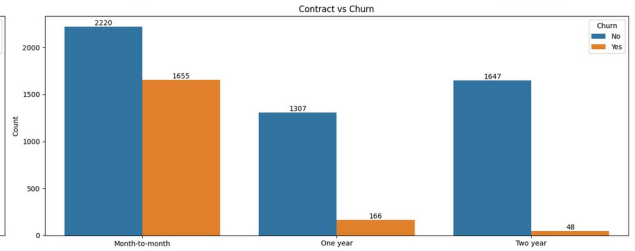
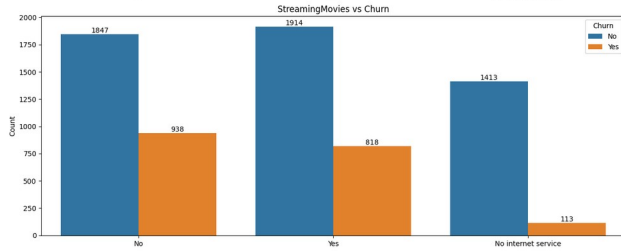
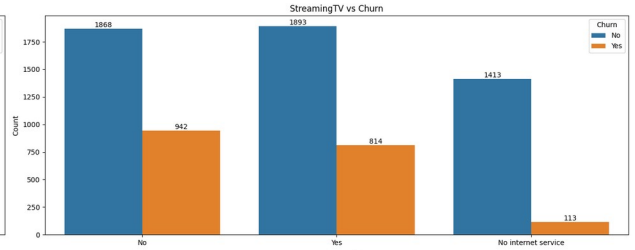
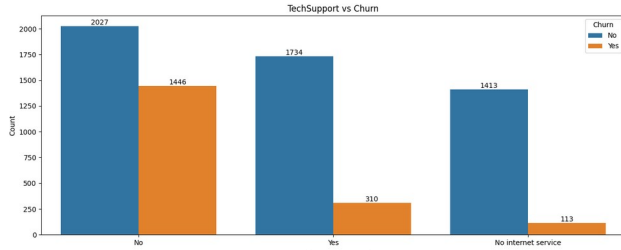
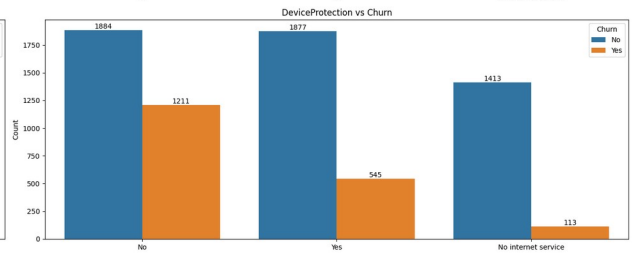
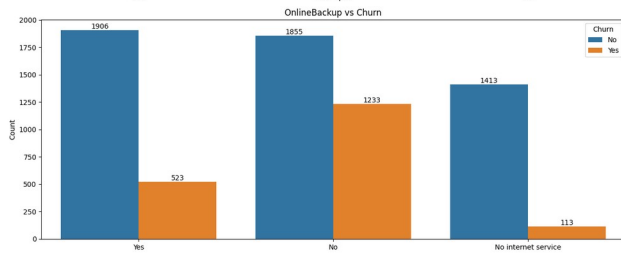
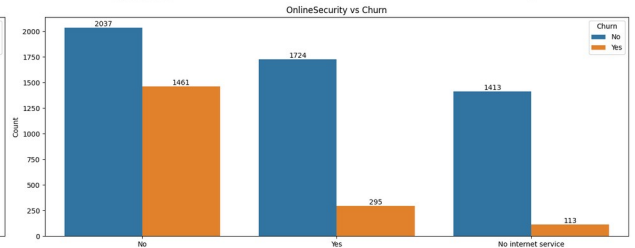
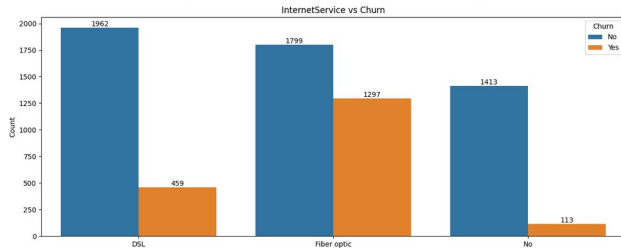
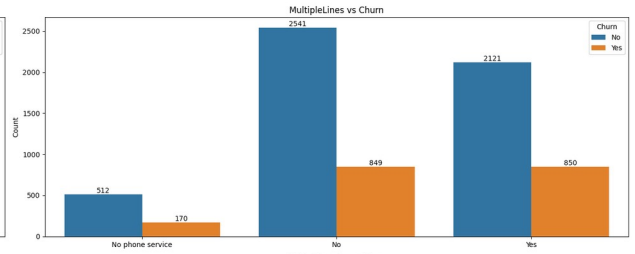
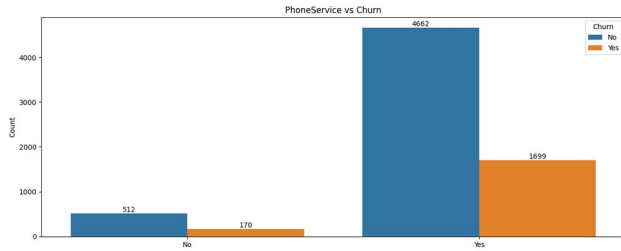


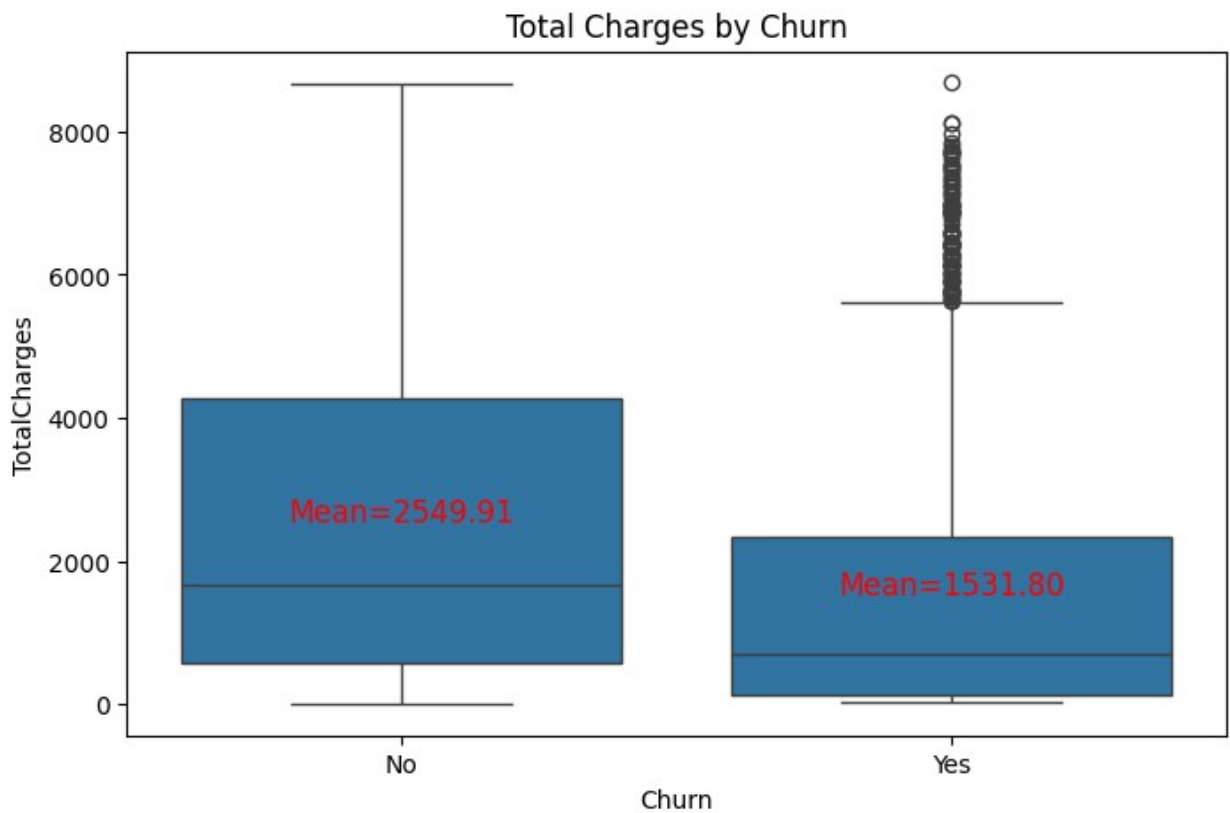
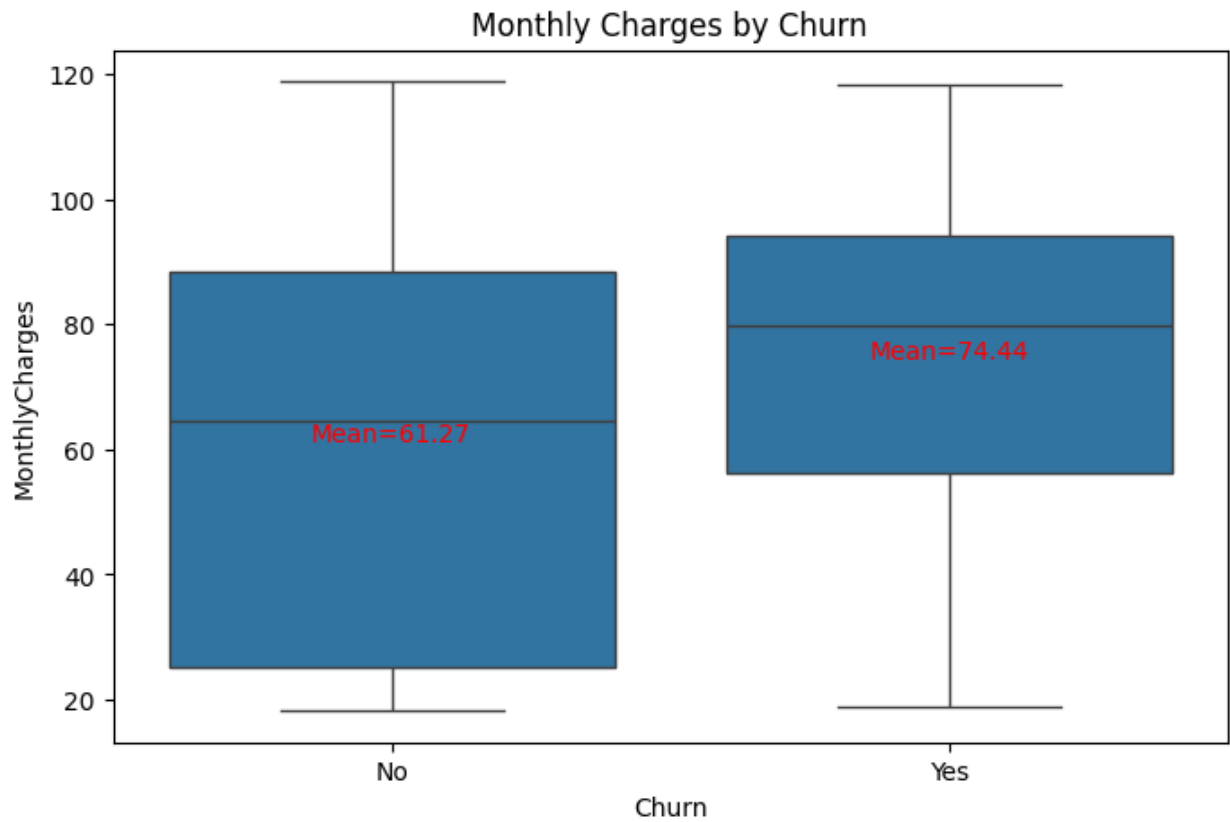


Percentage of Customers by SeniorCitizen










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<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
# Save cleaned dataset
df.to_csv(
    "Telecom_Customer_Churn_Cleaned.csv",
    index=False
)
files.download(
    "Telecom_Customer_Churn_Cleaned.csv"
)
<IPython.core.display.Javascript object>
<IPython.core.display.Javascript object>
```